COMPUTER ENGINEERING

QUEZON CITY

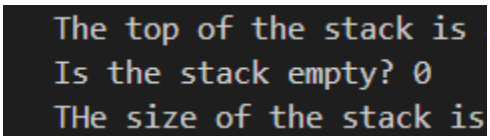
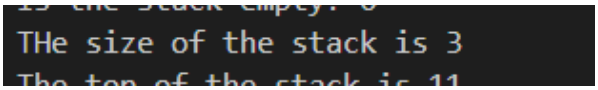
COURSE CODE/TITLE:	CPE 010 Data Structure and Algorithm	**SECTION:**	CPE21S2
DATE PERFORMED:	07/31/26	**DATE SUBMITTED:**	08/05/26
NAME:	Nieles, Cenndy M.	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. HOA4.1___ [Stacks]

1. PROCEDURE OUTPUT

What procedures did you perform?

ILOA

Screenshot	Observation
	The stack has elements in it at this point.
	The output show the stack size of 3 and the top element is 11 with the empty check is still returning 0 which is false.

The top of the stack1 is 12	The output has shown the separate stacks being referenced the stack 1 and stack2 with stack size
The top of the stack is 8 The top of the stack is 7	The top of the output shows the value of 8 and 7. This pattern is consistent with pop being called between the two Top

ILOB.

```
n.cpp
n.exe
n.cpp
n.exe
n.cpp
kList.h
n.cp
n.cpp
n.exe
Launcher.exe' '--stdin=Microsoft-MIEngine-In-rodptf3i.0kz' '--st
Engine-Pid-y3h0uipu.0jp' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe
================================================
Stack Operations
1. PUSH 2. POP 3. TOP 4. isEmpty 5. isFull
================================================
1
Enter a new value
4
================================================
Stack Operations
1. PUSH 2. POP 3. TOP 4. isEmpty 5. isFull
================================================
5
Is the stack full? 0
================================================
Stack Operations
1. PUSH 2. POP 3. TOP 4. isEmpty 5. isFull
================================================
3
Top Element: 4
================================================
Stack Operations
1. PUSH 2. POP 3. TOP 4. isEmpty 5. isFull
================================================
2
Popping: 4
================================================
Stack Operations
1. PUSH 2. POP 3. TOP 4. isEmpty 5. isFull
================================================
1
Enter a new value
4
================================================
Stack Operations
1. PUSH 2. POP 3. TOP 4. isEmpty 5. isFull
================================================
```

B.2. Stacks using Linked Lists

OUTPUT

```
PS C:\Users\joyni\OneDrive\Desktop\HOA4P1>  ${env:DEBUGINFOD_URLS}=''; & 'c:\Users\joyni
 '--stdin=Microsoft-MIEngine-In-ngzwnmtg.vcs' '--stdout=Microsoft-MIEngine-Out-eaphvdi0.
--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
After the first PUSH top of stack is: Top of Stack: 1
Stack elements (top to bottom): 1
After the second PUSH top of stack is: Top of Stack: 5
Stack elements (top to bottom): 5 -> 1
After the first POP operation, top of stack is: Top of Stack: 1
Stack elements (top to bottom): 1
After the second POP operation, top of stack is: Stack is Empty.
Stack is Empty.
Stack Underflow.
PS C:\Users\joyni\OneDrive\Desktop\HOA4P1>
```

SUPPLEMENTARY ACTIVITY

Expression	Valid(Y/N)	Output(Console Screenshoot)	Analysis
(A+B)+(C-D)	Y	msys64\ucrt64\bin\gdb.exe '--inte Expression: (A+B)+(C-D) Valid: Y	Every opening symbol has a matching closing symbol of the same type, and they close in correct (last-in-first-out)order
((A+B)+(C-D)	N	Expression: ((A+B)+(C-D) Valid: N	The three opening parentheses but only two closing parentheses.When the algorithm reaches the end of the of the string
((A+B)+[C-D])	Y	Expression: ((A+B)+[C-D]) Valid: Y	Different opening symbols are used -(,(, and [, then the final)
((A+B]+[C-D]}	N	Expression: ((A+B]+[C-D]} Valid: N	This fails because of mismatched symbol

			types,not missing ones

Array-Based Stack - Implementation to utilize a block of contiguous memory cells and an indextracks(top)It needs monitoring of stack overflow if the array is full and stack underflow.Usage & Memory Impact: Very memory-efficient implementation of a stack for situations when the size is known and does not change. Its efficiency comes from the lack of need to maintain pointers.

LinkedList - To allocate separate Node elements on the heap at run time.Operation Adding push elements at the beginning , and pop deleted head element by manual reassignment of the pointers.

C++ Standard Template Library - Manipulation and utilizing the code allocation of the memory, and all pointers manipulation is done automatically by the compiler.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.



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